📡 Ethernet Lan Switching Part 1 (10.23.2026)

1. OSI Model Context

- **Layer 1 (Physical):** Deals with the actual cables, connectors, and signals (electrical pulses, light).
- **Layer 2 (Data Link):** Deals with **Ethernet Frames** and **MAC Addressing**.
  - *Note:* The focus here is on the logical addressing (MAC) and how devices talk on the same local network.

2. The Ethernet Frame Structure

Understanding the frame is crucial for troubleshooting and understanding how switches operate.

Field	Size	Description
Preamble	7 Bytes	**Clock Synchronization.** A pattern of alternating 1s and 0s (10101010) used so the receiving device can synchronize its clock with the sender.
SFD (Start Frame Delimiter)	1 Byte	**Frame Start.** Ends with `10101011`. Tells the receiver "The next bits are the destination MAC address."
Destination MAC	6 Bytes	The intended recipient of the frame.
Source MAC	6 Bytes	The sender of the frame.
Type / Length	2 Bytes	Tells the receiver what protocol is inside (e.g., IPv4, IPv6, ARP).
Data / Payload	46 - 1500 Bytes	The actual encapsulated data (PDU from Layer 3).
FCS (Frame Check Sequence)	4 Bytes	**Error Detection.** Uses CRC (Cyclic Redundancy Check) to detect corrupted data.

> 💡 **Exam Tip:** The Preamble + SFD are often considered the "overhead" to prepare the receiver. If the FCS fails, the frame is **dropped**.

3. MAC Addresses (Media Access Control)

A MAC address is a unique identifier assigned to a network interface card (NIC).

- **Format:** Hexadecimal (Base 16).
- **Length: 48 bits** (6 bytes).
- **Notation:** Usually written as 12 hex digits (e.g., `E8BA.7011.2874`).

The Structure of a MAC Address

1. **OUI (Organizationally Unique Identifier):**
   - The **first 24 bits** (first half) of the address.
   - It identifies the **Manufacturer** (e.g., Cisco, Apple, Intel).
   - *Quiz Example:* In `E8BA.7011.2874`, the OUI is `E8BA.70`.
2. **Device Identifier:**
   - The **last 24 bits** (second half).
   - Unique to that specific device (assigned by the manufacturer).

Types of Addresses

- **Unicast:** A single specific device.
- **Broadcast:** All devices on the subnet (FF:FF:FF:FF:FF:FF).
- **Multicast:** A group of devices.

4. How Switches Operate (The MAC Address Table)

A switch is a Layer 2 device. Its main job is to forward frames based on the **Destination MAC Address**. To do this, it builds a **MAC Address Table** (also called CAM table).

The Learning Process (Quiz Question)

Question: Which field does a switch use to populate its MAC address table?
Answer: The **Source MAC Address**.

- **Why?** When a frame enters an interface, the switch looks at the **Source MAC** and thinks: *"Ah, this device is connected to this port. I will add it to my table."*

Forwarding Logic

1. **Known Unicast:**
    - The switch looks up the **Destination MAC** in its table.
    - If it finds a match, it **forwards** the frame out of the specific interface associated with that MAC.
2. **Unknown Unicast (Flooding):**
    - The switch looks up the **Destination MAC**.
    - If the MAC is **not** in the table, it **floods** the frame.
    - **Flooding:** The switch sends the frame out of **all interfaces except the one it was received on**.
    - *Note:* It does not forward out the interface it came from (that would be a loop).

> ⚠️ **Important Nuance:** The switch learns the Source MAC *before* it forwards the frame. So, if PC-A sends a frame to PC-B, and the switch doesn't know where PC-B is, it floods the frame. However, it *does* learn PC-A's location during this process.

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5. Quiz Summary & Answers

Here is a quick review of the quiz questions provided:

1. **Which field of an Ethernet frame provides receiver clock synchronization?**
    - ✅ **Preamble** (Allows the receiver to sync its clock).
2. **How long is the physical address of a network device?**
    - ✅ **48 bits** (Because the MAC address is 6 bytes).
3. **What is the OUI of this MAC address? E8BA.7011.2874**
    - ✅ **E8BA.70** (The first 24 bits / 3 bytes).
4. **Which field of an Ethernet frame does a switch use to populate its MAC address table?**
    - ✅ **Source MAC Address** (The switch learns which port the sender is on).
5. **What kind of frame does a switch flood out of all interfaces except the one it was received on?**
    - ✅ **Unknown Unicast** (The destination MAC is not yet in the MAC table).

📝 Mnemonic / Cheat Sheet

- **Preamble** = **P**hysical **P**rep (Sync).
- **Source MAC** = **S**witch **S**aves (Learning).
- **Destination MAC** = **D**irect **D**elivery (Forwarding).
- **Unknown** = **U**niversal Flood (Sending everywhere).

📊 Comparison: Hub vs. Switch (Context for Part 1)

Feature	Hub (Layer 1)	Switch (Layer 2)
Traffic	Floods everything	Forwards intelligently
Table	None	MAC Address Table
Collision Domain	All ports share one	Each port is its own

Note: This video focused heavily on Switch logic, which is the foundation for understanding VLANs and Spanning Tree Protocol (STP) in later videos.